

Beam Quality (M^2) of the Transverse-Field Speckle Datasets

What the caustic measurements show, and how they bear on the transverse Faraday hypothesis — supporting analysis for Chapter 5

In one sentence. The magnets change the beam propagation ratio M^2 of both fibres by at most a few percent, while the same magnets decorrelate the speckle by up to 19% — and that contrast is the useful result: whatever the field is doing to the guided light, it is not moving significant optical power up or down the mode-group ladder, which is what a bending, stress or gross mode-scrambling artefact would have done.

1. What this document is

Every image folder in the speckle campaign is a caustic z-scan: a sequence of camera frames recorded as the camera translates through the focus of the beam leaving the fibre. Those scans were never analysed for beam quality. This report extracts the ISO 11146 beam propagation ratio M^2 from all of them and asks two questions that Chapter 5 currently leaves open:

(a) How full is each fibre? Section 5.3 attributes the core-size scaling of the decorrelation to modal volume, and states explicitly that a full account “would require tracking the actual excited mode-group distribution ... which was not measured directly in this campaign”. M^2 is a direct measurement of exactly that quantity, and it was sitting in the data already.

(b) Do the magnets redistribute modal power? If the observed speckle decorrelation were produced by the magnets mechanically perturbing the fibre, or by any mechanism that scatters power broadly across mode groups, the modal power distribution would broaden and M^2 would rise measurably. This is a falsification test of the kind catalogued in Section 4.6.3, using an observable completely independent of the Pearson correlation.

2. How M^2 was measured

Beam widths are ISO 11146 second-moment ($D4\sigma$) diameters. The frame is background-corrected from a border ring; secondary bright lobes (the second Wollaston arm, and reflections near the sensor border, present in 320 of the 1641 frames) are masked, because a lobe far from the centroid dominates a second moment even when it carries little power; moments are then integrated over an elliptical aperture of three times the $D4\sigma$ diameter, iterated to convergence. For each scan the caustic $d^2(z) = a + bz + cz^2$ is fitted and

$$M^2 = (\pi / 8\lambda) \cdot \sqrt{(4ac - b^2)}$$

The fit uses a 3σ robust residual cut; every rejected frame is recorded and marked on the plots. Across the 64 step-index scans the median fit R^2 is 0.9986 and no fit failed.

Constant	Value	Source	Effect if wrong
Wavelength λ	532 nm	Table 4.1 / Ch. 4	M^2 scales as $1/\lambda$
Stage step per frame	10 mm	supplied	M^2 scales linearly with it
Pixel pitch	3.45 μm	Sony IMX273 (Blackfly S)	M^2 scales as pitch^2

These three constants set the **absolute** scale only. They are common to every scan, so every comparison in this report — fibre against fibre, configuration against configuration, magnet count against magnet count — is unaffected by them. Only the fill fractions of Section 3 depend on the absolute scale being right.

3. What M^2 measures in a step-index multimode fibre

For a step-index fibre the guided modes organise into principal groups $\mu = 2p + l + 1$, and the highest group the fibre can support is $\mu_{\max} = V/2$. A fully filled fibre radiates with

$$M^2(\text{full}) = \pi a \text{ NA} / \lambda = V/2 = \mu_{\max}$$

so **M^2 is, to a good approximation, a direct read-out of the highest mode group carrying appreciable power.** That makes it the natural instrument for the modal-fill question. Using $\text{NA} = 0.20$ and $\lambda = 532 \text{ nm}$ from Table 4.1:

Fibre	V	Modes ($V^2/2$)	$\mu_{\max} = M^2 \text{ ceiling}$	Measured M^2	Fill
50 μm	59	1744	29.5	15.34 $\pm$ 0.20	52%
105 μm	124	7689	62.0	42.33 $\pm$ 0.25	68%

The mode counts reproduce Table 4.1 exactly (1744 and 7689), which confirms the V-numbers are consistent with the thesis parameters.

Result 1 — the modal fill, measured directly. At the launch condition used for these scans the 50 μm fibre is excited to about mode group 15 of its 30 available (52% of its ladder), and the 105 μm fibre to about group 42 of 62 (68%). The larger core is therefore filled to a higher fraction of its own ladder, not a lower one. Roughly 118 modes carry the light in the 50 μm fibre against 896 in the 105 μm — a ratio of about 7.6. That is larger than the $(105/50)^2 \approx 4.4$ ratio of total mode counts quoted in Section 5.3, precisely because the larger core is also filled to a higher fraction of its own ladder.

4. The measurements

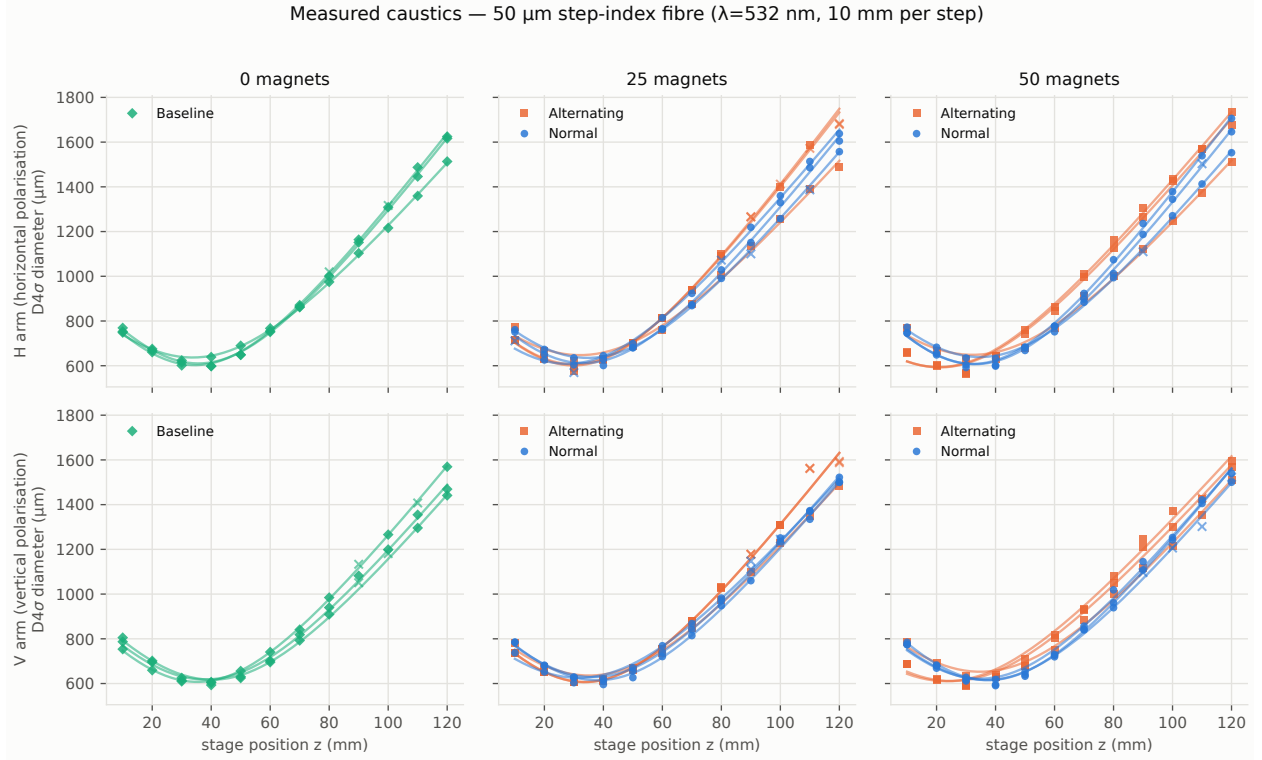


Figure 1. Measured caustics for the 50 μm step-index fibre. Each panel is one magnet count; rows are the two polarisation arms. Points are measured D4 σ diameters, curves are the fitted hyperbolae, and $\times$ marks a frame rejected by the robust residual cut. Repeat trials lie almost on top of one another, which is what makes the few-percent comparisons that follow meaningful. Note that the waist sits near the start of the travel — see the limitation in Section 8.

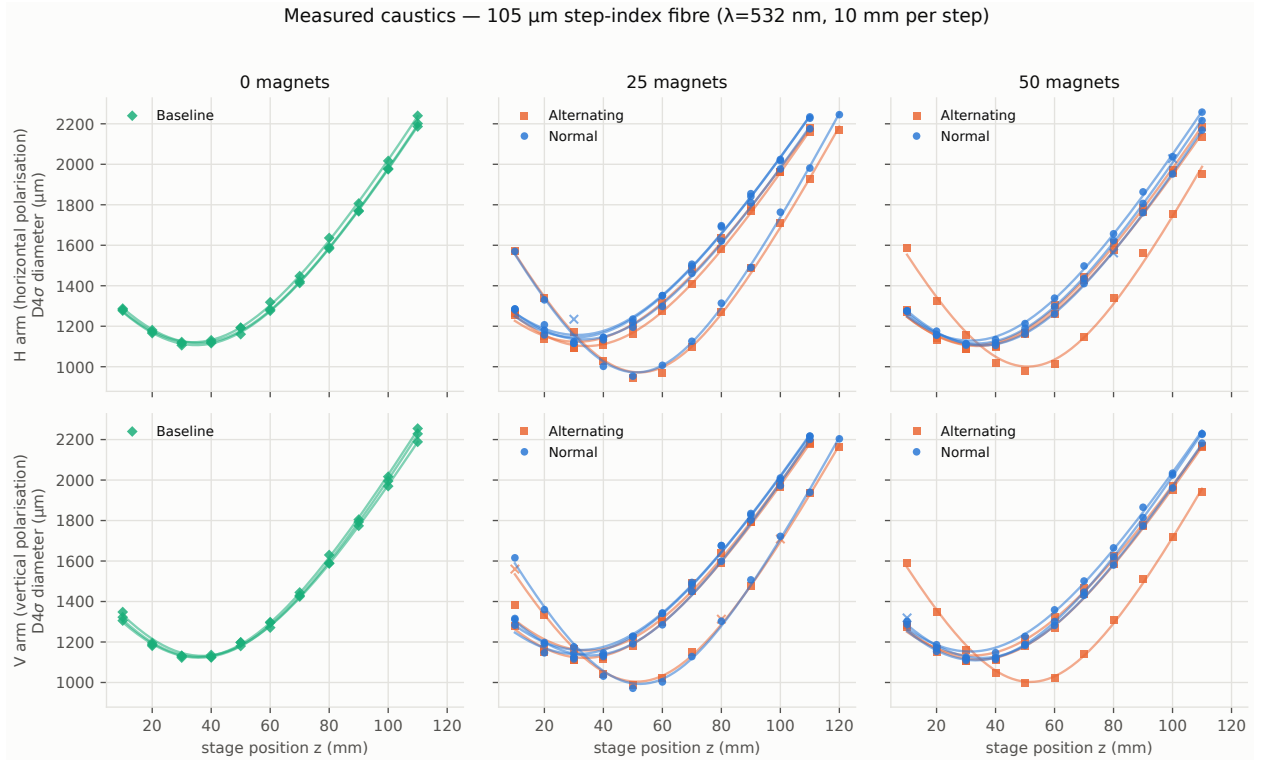


Figure 2. The same for the 105 μm fibre. The beam is visibly larger at every z , consistent with the larger modal volume. One session (the scans without a trial number in the folder name) sits at a shifted waist position with a slightly smaller waist — a different alignment day. Since M^2 is invariant to where the waist falls, those scans still return M^2 values consistent with the rest, and they are retained.

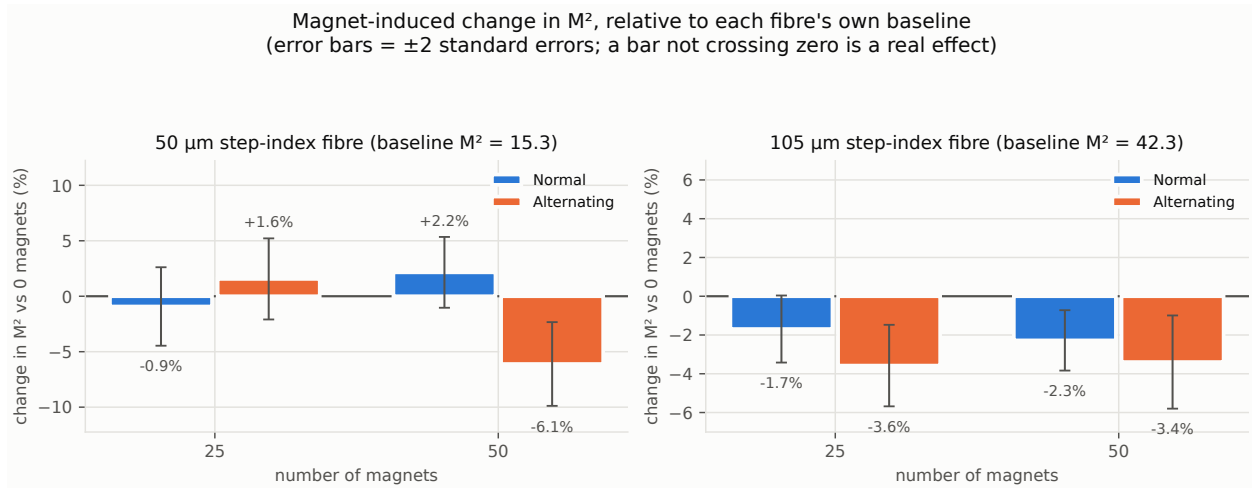


Figure 3. The headline result: change in M^2 relative to each fibre's own 0-magnet baseline, so that the few-percent effect under test is not hidden by the almost threefold difference in baseline between the fibres. Error bars are ± 2 standard errors, pooled over trials and over the two polarisation arms; a bar whose error interval does not cross zero is a real change. Every change that clears the noise is a reduction in M^2 .

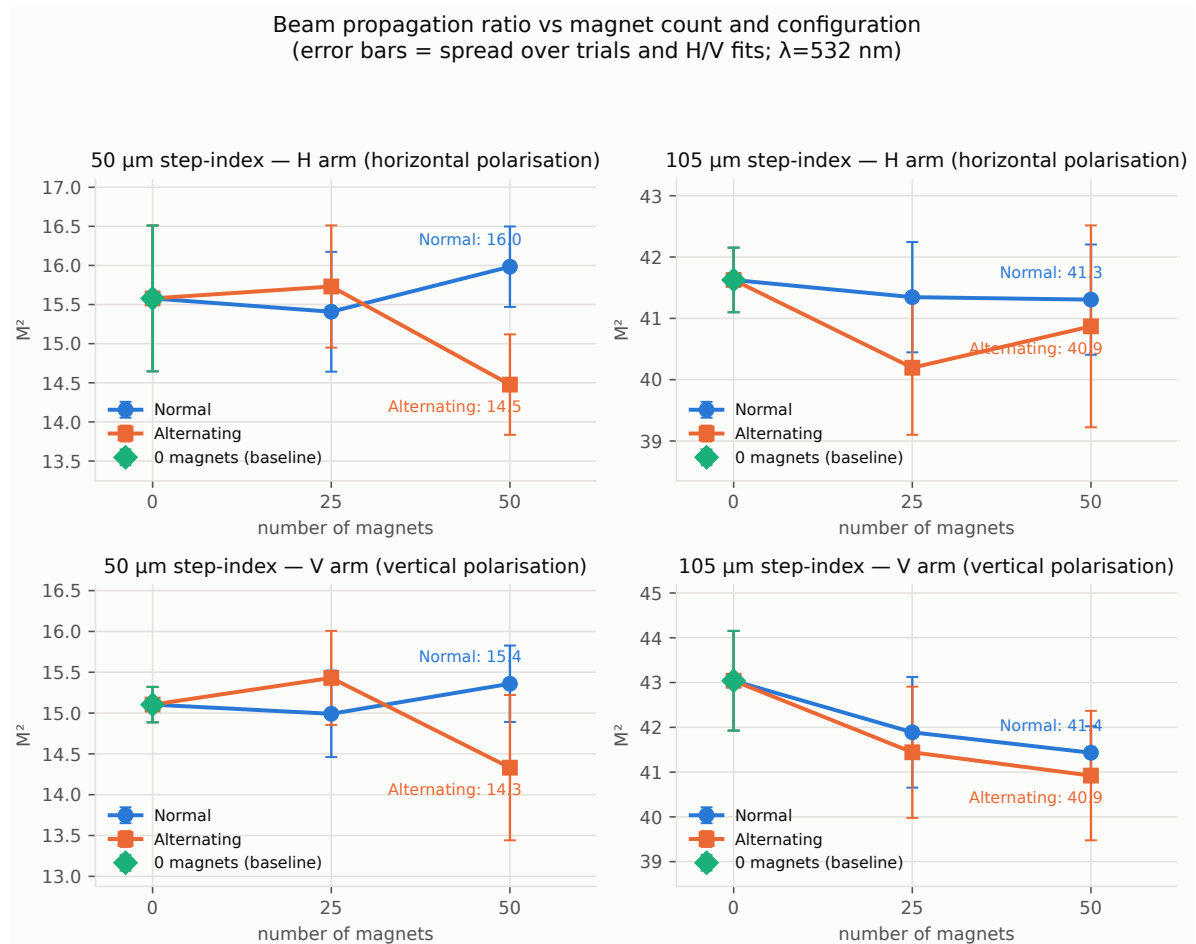


Figure 4. Absolute M^2 against magnet count, split by fibre and by polarisation arm, with the uniform (“Normal”) and alternating configurations overlaid. The 0-magnet baseline is shared by both configurations. The vertical scale spans only a few units of M^2 in every panel: on the scale of the baseline difference between the two fibres, all of these traces would be flat lines.

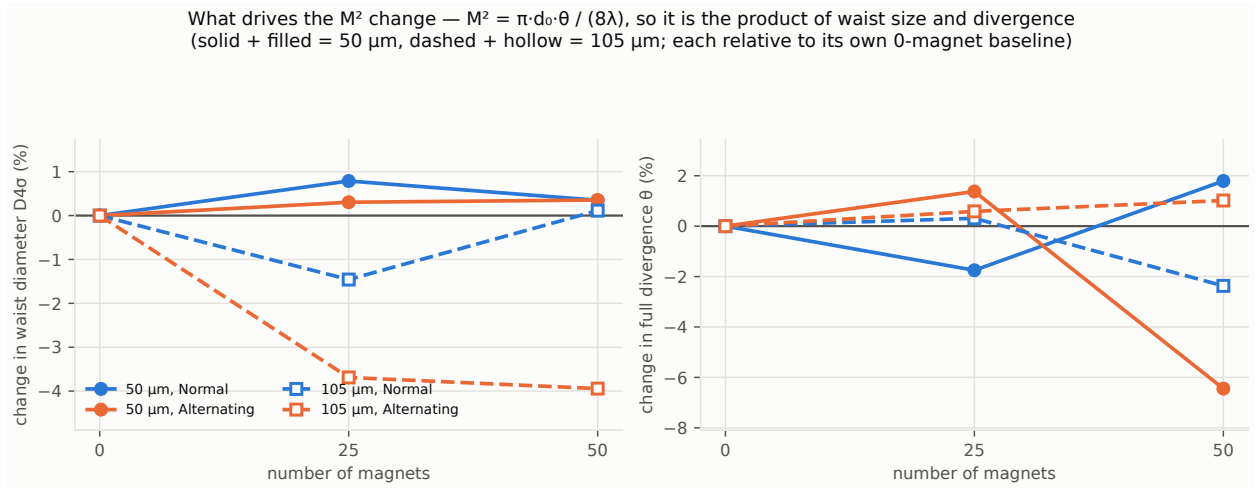


Figure 5. $M^2 = \pi \cdot d_0 \cdot \theta / (8\lambda)$, so any change in it is a change in the waist diameter, the far-field divergence, or both. Splitting the effect this way separates two different behaviours: in the 105 μm fibre the reduction comes mostly through a smaller waist, whereas the one large change in the 50 μm fibre (alternating, 50 magnets) comes through a lower divergence.

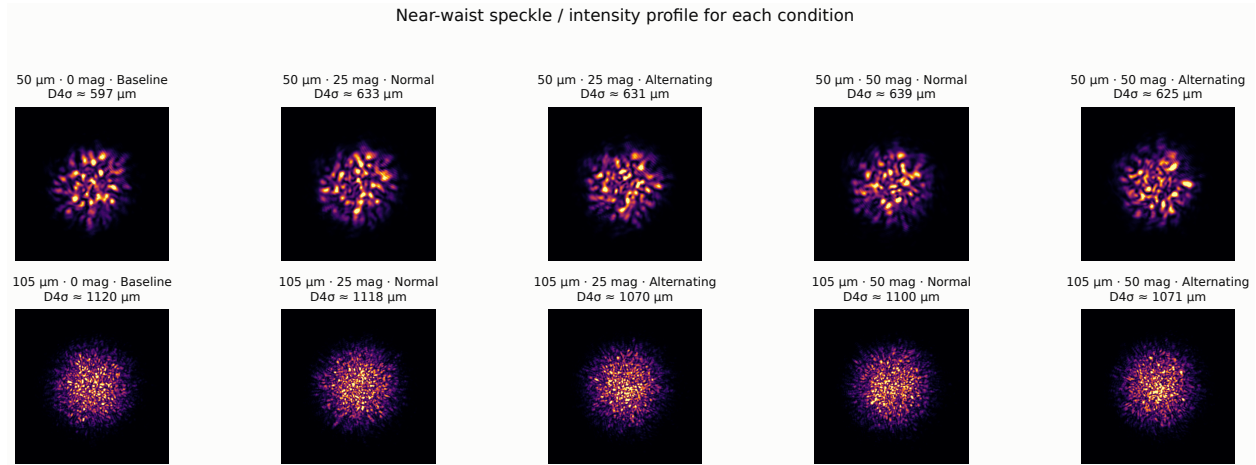


Figure 6. The near-waist intensity pattern for each condition, cropped to the beam and stretched to the 99.5th percentile. The 105 μm fibre carries visibly finer and far more numerous speckle grains than the 50 μm fibre — a direct visual counterpart of the mode counts in Section 3. Within a row the patterns are of similar size and grain, which is the M^2 result stated visually.

5. The numbers

Fibre	Configuration	Magnets	$M^2 \pm \text{s.e.}$	Δ vs baseline	% change	Clears noise?
50 μm	baseline (0 magnets)	0	15.34 ± 0.20	—	—	—
50 μm	uniform (non-alternating)	25	15.20 ± 0.19	-0.14	-0.9%	no
50 μm	uniform (non-alternating)	50	15.67 ± 0.14	+0.33	+2.2%	no
50 μm	alternating	25	15.58 ± 0.20	+0.24	+1.6%	no
50 μm	alternating	50	14.40 ± 0.22	-0.94	-6.1%	yes
105 μm	baseline (0 magnets)	0	42.33 ± 0.25	—	—	—
105 μm	uniform (non-alternating)	25	41.62 ± 0.27	-0.72	-1.7%	no
105 μm	uniform (non-alternating)	50	41.37 ± 0.22	-0.96	-2.3%	yes
105 μm	alternating	25	40.82 ± 0.37	-1.52	-3.6%	yes
105 μm	alternating	50	40.90 ± 0.45	-1.44	-3.4%	yes

“Clears noise” means the difference exceeds twice the combined standard error of the two means, a roughly 95% criterion. Each condition pools three trials and two polarisation arms.

6. What this means for the hypothesis

6.1 The decorrelation is not accompanied by mode-group redistribution

Chapter 5 reports speckle decorrelation reaching $1 - \rho \approx 0.18$ for the 50 μm fibre and 0.19 for the 105 μm fibre under the uniform configuration at maximum off-normal incidence. Over the same magnet range, M^2 moves by at most 6.1% and 3.6% respectively. The speckle pattern is being comprehensively rearranged while the modal power distribution across the group ladder stays put.

That dissociation is exactly what a **phase-scrambling, power-conserving** interaction looks like. Speckle correlation responds to the relative phases of the modes; M^2 does not — it responds only to how much power sits in which mode group. An interaction that reshuffles amplitude and phase among modes of the same principal group changes the speckle completely and leaves M^2 untouched, because modes within one group share the same μ and therefore make the same contribution to the space-angle second moment. This is precisely the near-degenerate intra-group vector channel identified in Section 3.6 as the pathway capable of producing the measured effect.

Result 2. The data are consistent with the intra-group near-degenerate channel of Section 3.6 and give it independent support from an observable that has nothing to do with the Pearson correlation. **They do not, however, prove it** — see the sensitivity argument immediately below, which is the honest limit of what M^2 can settle.

6.2 How much power transfer M^2 can actually exclude

This needs stating carefully, because it is where the argument could be overclaimed. If a fraction f of the power is moved by $\Delta\mu$ groups, then to first order $\Delta M^2/M^2 \approx f\Delta\mu/\mu$. The sensitivity therefore depends strongly on how far the power moves:

Fibre	M^2 resolution (2 s.e.)	Bound on f for $\Delta\mu = 1$	Bound on f for broadband scattering
50 μm	2.6% (± 0.39 in μ)	$f \leq 39\%$	$f \leq 6\%$
105 μm	1.2% (± 0.50 in μ)	$f \leq 50\%$	$f \leq 3\%$

So M^2 is a **strong** constraint on broadband redistribution — power scattered widely across the ladder, as bending or gross mode mixing would do — bounding it at a few percent of the total. It is a **weak** constraint on nearest-neighbour transfer, and the inter-group channel of Section 3.4 has $|\Delta l| = 1$ and hence $|\Delta\mu| = 1$ as its minimum step. M^2 therefore cannot by itself distinguish the inter-group channel from the intra-group one. What it does do is close off the mechanical and gross-scattering alternatives, which is its proper role in the falsification programme of Section 4.6.3. It also cannot test the 2×10^{-7} ceiling of Eq. 3.47: that is some five orders of magnitude below anything this measurement can resolve.

6.3 A further falsification of the mechanical pathway

Section 5.1 already separates mechanical disturbance from magneto-optic coupling by its temporal signature — a step discontinuity against a smooth decay. The M^2 data close the same door from a different side. Macrobending and clamping stress do two things to a multimode fibre: they couple power between mode groups, and they strip the highest-order groups. Both change M^2 , and at the field strengths and handling involved in fitting 50 magnet pairs they would change it by far more than the few percent seen here. The near-constancy of M^2 across 0, 25 and 50 magnets is therefore independent evidence that mounting the magnet array is not mechanically reworking the fibre's modal content.

6.4 The small reductions, and how far to push them

Four of the eight magnet conditions show a change that clears the noise, and all four are reductions in M^2 . The 105 μm fibre is the consistent one: three of its four conditions move, all downward, by 2-4%, and Figure 5 shows the reduction arriving through a smaller waist. Two readings are available and the data do not separate them. Either the magnet array is acting as a very weak mode filter — slight bending along the rail preferentially attenuating the least-confined highest-order modes, which are the ones that contribute most to the second moment — or the launch drifted very slightly over the sequence in which the magnets were fitted. The second is entirely plausible at this magnitude. What matters for the hypothesis is that neither reading involves power being scattered up the ladder, which is what a strong inter-group coupling mechanism would have produced.

6.5 The uniform-versus-alternating contrast

Section 4.6.3.3 uses the uniform-versus-alternating contrast as the discriminator between a field-linear effect and a B^2 effect. In the decorrelation data that contrast is large and is the central result. In the M^2 data it is mostly absent, as the power-conservation argument above predicts. For the 105 μm fibre the two configurations differ by 0.47 in M^2 at 50 magnets, within the combined standard error, so they are indistinguishable. For the 50 μm fibre they do differ — uniform 15.67 against alternating 14.40 — but this rests on a single condition in the noisier of the two fibres, and it is the alternating configuration that sits lower, which is not the ordering the transverse Faraday mechanism would predict if M^2 were tracking the coupling strength. It is better treated as an outlier to be re-measured than as a result.

7. The same thing in plain words

M^2 is a single number for how good a laser beam is. $M^2 = 1$ is a perfect beam; the bigger the number, the messier and faster-spreading the beam. A multimode fibre scrambles light into many “modes”, so its output has a large M^2 — and crucially, that number tells you **how many** of the fibre's modes are carrying light.

Two things came out of measuring it.

First, a number the thesis says it did not have. Chapter 5 argues the 105 μm fibre reacts more strongly than the 50 μm one because it has more modes, but notes the actual modal content was never measured. It has now been measured, from images already taken: the 50 μm fibre was running at about half of its mode capacity and the 105 μm at about two thirds, carrying roughly 120 and 900 modes respectively. That is a real, quotable number for the modal-volume argument.

Second, and more important: the magnets barely change the beam at all. The speckle pattern changes a lot when the magnets are on — that is the whole result of the thesis — but the beam's M^2 stays essentially the same. Think of a choir. The speckle pattern is the sound of the choir, which changes completely if the singers change who is singing which note. M^2 is the range of the choir — how low the basses and how high the sopranos go. The measurement says the singers are swapping notes among themselves, but nobody is singing outside the range they started in. Something is rearranging the light, and it is not simply shoving it into higher-order modes.

That matters because the most boring explanation for the whole effect — “the magnets are just bending or squeezing the fibre” — would have shown up here. Bending a multimode fibre does change its M^2 , and noticeably. It did not change. So this is one more alternative explanation closed off, using a measurement that has nothing to do with the correlation numbers the rest of the chapter relies on.

8. Does it support the hypothesis?

Yes, but as supporting and falsifying evidence, not as proof.

It supports the project in three ways. It supplies the direct measurement of modal fill that Section 5.3 identifies as missing, and the numbers back the modal-volume reading of the core-size scaling. It shows the decorrelation happens without power moving across the mode-group ladder, which is the behaviour expected of the near-degenerate intra-group channel proposed in Section 3.6, from an observable independent of the Pearson correlation. And it independently closes the mechanical and gross-scattering alternatives that Section 4.6.3 sets out to falsify.

It does not prove the transverse Faraday mechanism. M^2 is a null result here: it says what is not happening. It cannot separate the inter-group channel from the intra-group one, because both can operate at $|\Delta\mu| = 1$, where M^2 is insensitive. It is nowhere near sensitive enough to test the theoretical ceiling of Eq. 3.47. And it shows no uniform-versus-alternating asymmetry, so it adds nothing to the field-linearity argument — which, on the power-conservation reasoning of Section 6.1, is what it should do.

9. Limitations you should state if you use this

Limitation	What to say
Launch condition not recorded	The folder names for the caustic scans do not record which incidence configuration (I, II or III of Section 5.3) was in use. The fill fractions in Section 3 are therefore valid for “the launch used for these scans”, not for a named configuration. If you know which it was, say so; if the scans span more than one, the fill numbers should be quoted per configuration.
Fill fractions depend on the absolute calibration	Unlike everything else here, the 52% and 68% figures move if the 10 mm stage step or 532 nm wavelength is wrong. The comparisons and the null result do not.
ISO 11146 sampling is not met	In every scan the waist sits near the start of the travel, so there are only about three frames before it and the scan reaches roughly two Rayleigh lengths beyond. The standard asks for at least five on each side. Fits are excellent (median $R^2 = 0.9986$) and reproducible, so relative comparisons hold, but the absolute M^2 carries more systematic uncertainty than the quoted standard errors.
Three trials, not ten	The caustic scans have $n = 3$ per condition against the $n = 10$ of the decorrelation protocol. The error bars reflect this.
A single significant 50 μm result	The -6.1% at 50 alternating magnets rests on one condition and should be re-measured before being leaned on.

10. Suggested placement in Chapter 5

The material divides naturally in two, matching the two questions in Section 1 above.

Content	Suggested home
Modal fill: V-numbers, ceilings, measured M^2 , fill fractions, mode counts (Section 3, Figure 6)	Into 5.3 , at the “Core size and modal volume” paragraph, replacing the sentence conceding that the excited mode-group distribution was not measured.

Content	Suggested home
Power-conservation falsification: Figures 3 and 5, the numbers table, and Sections 6.1–6.4	As a new section after 5.5 , e.g. “5.6 Modal Power Conservation under the Applied Field”, renumbering the regime diagnostic and theory confrontation that follow.
The intra-group reading (Section 6.1)	Cross-reference from 5.7.3 , “Evidence Bearing on the Resolution”, which is where the case for the Section 3.6 channel is assembled.
Figures 1, 2 and 4	An appendix, or supporting figures for the new section. They document the measurement rather than carry the argument.

One caution on wording. Section 5.3 explains the convergence of the two fibres at maximum off-normal incidence by suggesting a given launch populates a larger fraction of the 50 μm fibre's ladder than of the 105 μm fibre's. At the launch used for the caustic scans the measured fill runs the other way (52% against 68%). This is not necessarily a contradiction — the launch conditions may differ — but the two statements should not sit in the same chapter unreconciled. Either establish which configuration the caustic scans correspond to, or soften the fractional-fill argument to a possibility rather than an explanation.

11. Provenance

All numbers and figures come from `m2_analysis.py` in the project repository, run over the 1641 frames in 98 z-scans under `M-Squared Results/`. Per-frame widths, per-scan fits and the condition summary are in `analysis_output/` as CSV, including every quality flag and every rejected frame, so any figure here can be traced back to individual images. This PDF is generated by `make_m2_report_pdf.py` and reads those CSVs directly, so the prose cannot drift from the data. Every figure is written as PDF, SVG and PNG; the figures embedded above are the vector PDFs, so they stay sharp at any zoom and can be dropped straight into a LaTeX document with `\includegraphics`.